

Devkrishna U S

+91 9447245274 — devkrishnaus@gmail.com — linkedin.com — github.com

Professional Summary

Highly motivated Computer Science undergraduate at VIT Bhopal specializing in **Cybersecurity**. Hands-on experience in vulnerability assessment and penetration testing (VAPT), network security, and building system-wide defense suites. Skilled in analyzing malicious network traffic and exploit vectors while applying frameworks such as the OWASP Top 10, NIST Cybersecurity Framework, and ISO/IEC 27001. Seeking to leverage a strong offensive security mindset and hands-on tool expertise in Ethical Hacking and VAPT.

Education

VIT Bhopal University	Bhopal, India
B.Tech – Computer Science & Engineering (<i>Cybersecurity</i>)	2023 – 2027

Technical Skills

- **Offensive Security & VAPT:** Ethical Hacking, Vulnerability Assessment and Penetration Testing (VAPT), Web Application Security (OWASP Top 10), Network Security, Reconnaissance, Threat Analysis, Incident Response
- **Security Tools:** Burp Suite, Nmap, Wireshark, Metasploit, Hashcat, Kali Linux
- **Frameworks & Compliance:** NIST Cybersecurity Framework, ISO/IEC 27001, OWASP Top 10, Data Privacy Techniques, Risk Assessment
- **Programming & Scripting:** Python, Java
- **Version Control & DevOps:** Git, GitHub Operations
- **Operating Systems:** Linux (Kali), Windows

Projects

Network Intrusion Detection System

Tech Stack: Python, Flask, Scapy, SQLite, Socket.IO, ReportLab

- Engineered a real-time network intrusion detection system capable of simultaneous multi-interface packet capture to detect port scans, SYN/ICMP floods, SQL injection, and command injection attacks.
- Developed a reactive Flask web dashboard featuring live WebSocket-driven attack feeds, dynamic analytics charts, and a SQLite persistence layer for comprehensive audit logging.
- Integrated an automated alerting pipeline with Gmail SMTP for instant HTML notifications on critical threats, alongside executive-grade PDF security report generation.

Image Steganography & Authenticated Encryption Platform

Tech Stack: Python, FastAPI, React, Vite, Tailwind CSS, AES-256-GCM

- Architected a full-stack secure communication platform integrating AES-256-GCM authenticated encryption with Adaptive LSB steganography to securely embed payloads within PNG and BMP images.
- Implemented military-grade brute-force resistance utilizing PBKDF2-HMAC-SHA256 (600,000 iterations) alongside an anti-side-channel error model to thwart unauthorized payload extraction.
- Engineered an asynchronous FastAPI backend to process high-resolution files and calculate real-time visual quality metrics (achieving PSNR > 51 dB and SSIM > 0.999), connected to a reactive React frontend.

Autonomous AI Business & Intelligence Analyst Platform

Tech Stack: Python, FastAPI, React 19, TypeScript, Gemini API, Scikit-Learn, Plotly

- Architected and deployed an autonomous AI-driven BI platform featuring conversational data querying, automated exploratory analysis (EDA), predictive forecasting, and executive reporting.
- Implemented a hybrid Natural Language Query (NLQ) engine utilizing Google Gemini API and semantic heuristics to translate plain English into optimized Pandas queries with evidence citations and sub-second execution.
- Built a predictive analytics suite incorporating Isolation Forest for multi-dimensional risk/outlier scoring and Random Forest Regressor for automated feature importance driver analysis.
- Developed a responsive single-page application using React, TypeScript, and Tailwind CSS, providing smooth file ingestion, interactive Plotly charts, scenario simulation sliders, and one-click PDF export.

Certifications & Professional Development

- **AWS Certified Solutions Architect – Associate**
- **Google IT Support Professional Certificate**
- **ServiceNow System Administration Fundamentals – On-Demand Course**